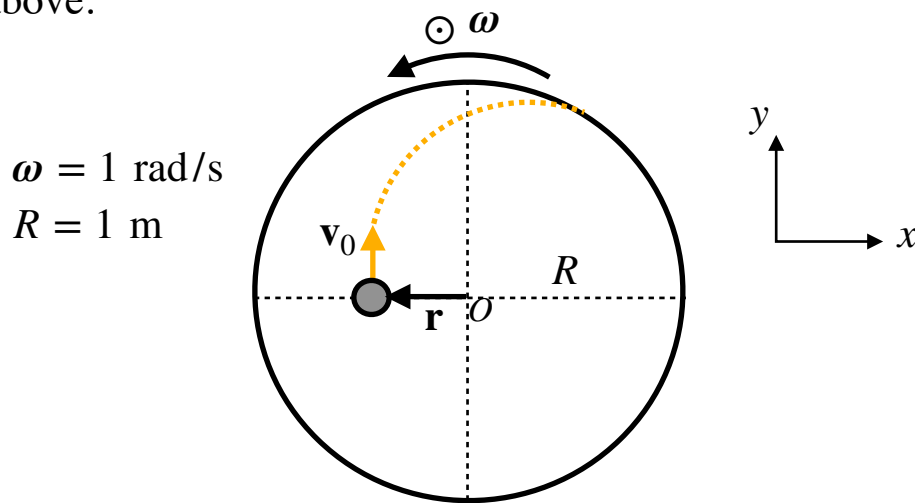


**Advanced Classical Mechanics (PHYS 615)**  
**Extra Credit Project 3 (due on date of Exam I)**

**Fall 2026**

The points obtained in this project will be added towards Exam I. To obtain full credit, your computer source code + calculations done on paper **MUST** be provided along with the plots.

A student is performing measurements with a hockey puck on a large merry-go-round with a smooth (frictionless) horizontal, flat surface. The merry-go-round has a constant angular velocity  $\boldsymbol{\omega} = \omega \hat{z}$  and rotates counterclockwise as seen from above.



- (a) **(5 pts)** The effective acceleration on the puck (as observed in the rotating frame) after it is given an initial push is:

$$\mathbf{a}_{\text{eff}} = \frac{\mathbf{F}_{\text{eff}}}{m} = - \underbrace{\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})}_{\text{centrifugal term}} - \underbrace{2\boldsymbol{\omega} \times \mathbf{v}}_{\text{coriolis term}}$$

where  $\mathbf{r} = x\hat{x} + y\hat{y}$  and  $\mathbf{v} = v_x\hat{x} + v_y\hat{y}$  an arbitrary position and velocity of the puck some time  $t > 0$  after it is given an initial push.

Determine the x- and y- components of the effective acceleration ( $a_x = dv_x/dt$ ,  $a_y = dv_y/dt$ ).

- (b) **(10 pts)** The equations in (a) are coupled differential equations for which there is no exact solutions and numerical methods must be used. Use Euler's method (introduced in Project #2) to solve the equations for  $[v_x(t), v_y(t)]$  and  $[x(t), y(t)]$

The velocity and position:

$$\mathbf{v}_{\text{eff}} = \int \mathbf{a}_{\text{eff}} dt, \quad \mathbf{r}_{\text{eff}} = \int \mathbf{v}_{\text{eff}} dt$$

are given by numerical integration:

$$v_{x,n+1} = v_{x,n} + a_{x,n} \Delta t$$

$$v_{y,n+1} = v_{y,n} + a_{y,n} \Delta t$$

$$x_{n+1} = x_n + v_{x,n} \Delta t$$

$$y_{n+1} = y_n + v_{y,n} \Delta t$$

where  $n$  represents number of step-sizes and  $\Delta t = t_{n+1} - t_n$  is the time step-size required to advance the numerical integrations

Perform the numerical calculation to determine the motion  $[x(t), y(t)]$  and make separate plots of  $y(t)$  vs.  $x(t)$  to show the different paths of the puck for the following initial conditions:

(i) initial condition ( $t = 0$ ):

$$(x_0, y_0) = (-0.5R, 0)$$

$$(v_{x,0}, v_{y,0}) = (0, 1.5 \text{ m/s})$$

(ii) initial condition ( $t = 0$ ):

$$(x_0, y_0) = (-0.5R, 0)$$

$$(v_{x,0}, v_{y,0}) = (0, 0.8 \text{ m/s})$$

(iii) initial condition ( $t = 0$ ):

$$(x_0, y_0) = (-0.5R, 0)$$

$$(v_{x,0}, v_{y,0}) = (0, 0.45 \text{ m/s})$$

(iv) initial condition ( $t = 0$ ):

$$(x_0, y_0) = (-0.5R, 0)$$

$$(v_{x,0}, v_{y,0}) = (0, 0.328 \text{ m/s})$$

(v) initial condition ( $t = 0$ ):

$$(x_0, y_0) = (-0.5R, 0)$$

$$(v_{x,0}, v_{y,0}) = (0.47 \cos(45^\circ), 0.47 \sin(45^\circ)) \text{ m/s}$$

(vi) initial condition ( $t = 0$ ):

$$(x_0, y_0) = (-0.5R, 0)$$

$$(v_{x,0}, v_{y,0}) = (0.283 \cos(45^\circ), 0.283 \sin(45^\circ)) \text{ m/s}$$

where (v) and (vi) have initial velocities set at an angle of 45 deg. relative to the x-axis, which will have more interesting trajectories.

Set the following constants to:  $\omega = 1 \text{ rad/sec}$ ,  $R = 1 \text{ m}$

When plotting the trajectories, also draw a circle of radius  $R$  to indicate the boundary of the merry-go-round, which will help visualize the trajectory of the hockey puck as well as when it goes out of bounds. If possible, you may also find the time  $T$  the puck takes to reach the boundary.